AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# TM240

**Introduction to computer graphics**

## Final Examination
## Fall Semester 2024/2025

## Date: 9 /Jan./ 2025

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| **Number of Exam Pages:** 4 (including this cover sheet) | **Time Allowed:** 120 minutes |

## Instructions:

- Total Marks: **100**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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**Part I:** This part consists of 10 multiple choice questions. For each question, encircle the label a, b, c, or d as correct answer. Each question carries 2 marks.

1 .....is a set theoretic function used to construct the union of two areas.

A add B exclusiveor
C union D interset

2 The ..... defines the pattern of opaque and transparent sections applied along the length of the line.

A ending B joins
C dash phase D dash style

3 In the class .....there is no separation between modeling and rendering

A J2D B Graphics
C Graphics2D D None of the above

4 The data of the filter that is commonly used to detect objects' edges in an image is .....

A {0.11,0.11,0.11,0.11,0.11,0.11,0.11,0.11,0.11} B {0f, -1f, 0f, -1f, 4f, -1f, 0f, -1f, 0f}
C {0f, -1f, 0f, -1f, 5f, -1f, 0f, -1f, 0f} D None of the above

5 .....are geometric transformations applied to objects to achieve proper placement of objects in the scene.

A modeler B Object transformation
C World transformation D Viewing transformation

6 ..... introduces dynamic changes to graphics contents and often creates a visual effect of motion

A Animation B None of the above
C None of the above D None of the above

7 In the .....winding rule the point drawing order is not required.

A Even-odd B Close
C Non-zero D None of the above

8 The function drawOval is an example of ..... functions

A Modeling and rendering B Rendering
C Modeling D None of the above

9 the function .....can be used to alter the geometry of a given image

A ConvolveOp B AffineTransformOp
C ColorConvertOp D LookupOp

10 .....function is used to concatenate transformation to the right of the current transformation matrix.

A strcat B preConcatenate
C concatenate D cat

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**Part II:** This part consists of four questions with a total of 80 Marks, solve all questions.

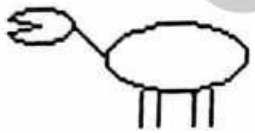
## Question 2 [20 Marks]

a. Describe the different types of affine transforms and how they are used in computer graphics.

b. Having the following code in Java:

```
Shape neck=new Line2D.Double(171,302,161,292);
Shape head=new Arc2D.Double(141,287,20,10,200,320,Arc2D.PIE);
g.draw(neck);
g.draw(head);
Shape lArm=new Line2D.Double(181,312,181,322);
g.draw(lArm);
Shape rArm=new Line2D.Double(186,312,186,322);
g.draw(rArm);
Ellipse2D body=new Ellipse2D.Double(171,292,40,20);
g.draw(body);
Shape Rleg=new Line2D.Double(201,312,201,322);
g.draw(Rleg);
Shape LLeg=new Line2D.Double(196,312,196,322);
g.draw(LLeg);
```

The code has the following out:



Modify the above code to output the following image adopting appropriate geometric transformations:



## Question 3 [20 Marks]

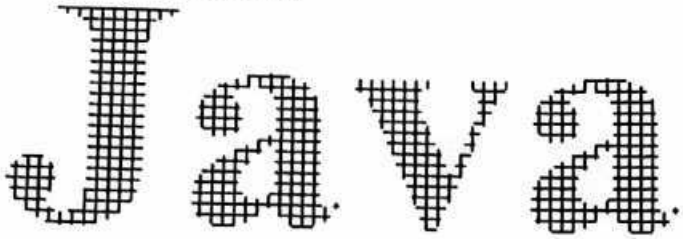
a. Explain the purpose and usage of the `GeneralPath` class and the different methods used to create and manipulate paths in Java2D.

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b. Having the following shape:



Write a java code segment to plot it.

## Question 4 [20 Marks]

a. Assuming that we have a number x that is represented in 15 bits, write down appropriate java code segment to divide this number into 3 colors (red, green and blue) with each color component of one byte. **[4 marks]**

b. Having an image object of type BufferedImage that has the following artifacts:

I. image is sheared towards the positive x-directions
II. image is sheared towards the negative x-directions
III. image is sheared towards the negative y-directions
IV. image is rotated clockwise around its center
V. image has inverted colors
VI. image is zoomed out
VII. image is to be written to the hard disk
VIII. A different image is to be loaded from the hard disk

Write a java code segment to resolve the above problems.

## Question 5 [20 marks]

a. Explain the limitations of Bézier curves for higher-degree polynomials and why they are not suitable for large sets of control points.**[5 marks]**

b. Given the following list of control points: **[15 marks]**
(0,38)
(37,30)

I. Derive the parametric Bezier equation for the curve controlled by the above point list.
II. Write a code to implement the parametric equations that you derived in the previous point.

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**End of questions**

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